Estimating turbulent broadening of cloud droplet size distributions

Recovering an effective diffusivity D_x from droplet-size histograms

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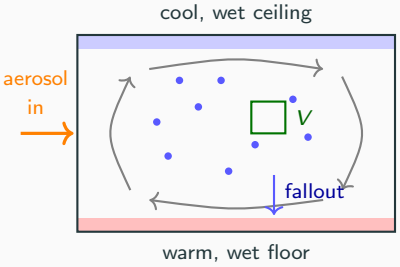
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The problem

Why droplet size distributions matter

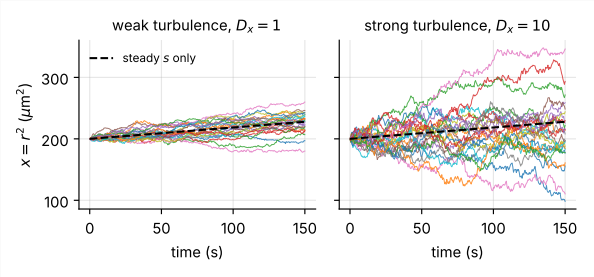
- The **width** of the droplet size distribution (DSD) controls how often droplets collide, and so when drizzle forms
- **Turbulence broadens the DSD**: droplets wander through moister and drier air and grow at different rates
- **Pi Chamber** (MTU): a steady turbulent cloud. Aerosol is injected, activates, grows, and falls out
- Clean clouds (few droplets) have broader DSDs than polluted ones (Chandrakar et al. 2016, 2018)

Question: how strong is the turbulent broadening, and can we infer it from measured DSDs?



Probe samples a small volume V.

Droplet growth as a random walk in $x = r^2$



25 simulated droplets starting at the same size.

- Condensation: $r\,\dfrac{dr}{dt} = \xi s \;\Rightarrow\; \dfrac{d(r^2)}{dt} = a\,s$, a constant rate in $x = r^2$
- Turbulence makes s fluctuate, so growth is a random walk
- Many droplets: **drift** (mean growth, Köhler effects) plus **diffusion** with coefficient D_x
- Heuristically $D_x \approx a^2 \sigma_s^2 \tau_s$ (variance and correlation time of s seen by a droplet)

D_x is an *effective* coefficient: it absorbs any broadening the drift does not explain.

Goal and roadmap

Goal: estimate a constant D_x from histograms of droplet sizes, with an honest error bar.

A ladder of synthetic experiments, adding one complication at a time; the true D_x is known, so we can check the answers.

1.	Steady state, exact data, only D_x unknown	**done**
2.	Steady state, instrument-like sampled data: Poisson counts, detection limit	**done**
3.	Truth from a finer grid, then from a stochastic particle model	next
4.	Estimate the supersaturation s together with D_x	next
5.	Simulation output (C-ODT, LES, DNS), then Pi Chamber data; time dependence	later

The forward model

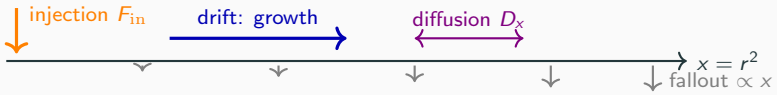
The steady Fokker–Planck equation

$$\frac{dF}{dx} + \lambda(x)\, N = 0, \qquad F = A(x)\, N - D_x \frac{dN}{dx}$$

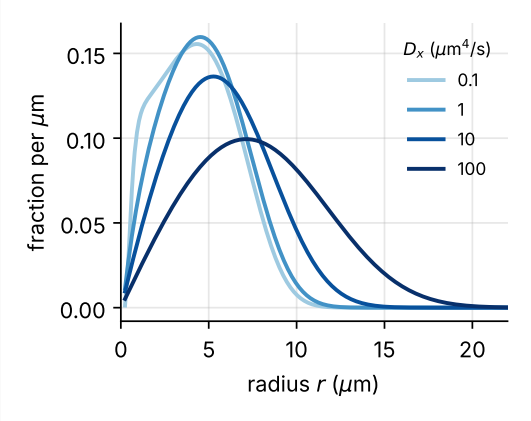
- F: flux of droplets growing past size x
- Drift $A = a\,(s - A_k/r + B_k/r^3)$: condensation with Köhler curvature and salt terms
- Fallout $\lambda = (C_{\mathrm{stk}}/h)\, x$: Stokes fall speed over chamber height
- Injection $F(x_{\min}) = F_{\mathrm{in}}$; no flux at $r = 30\,\mu\mathrm{m}$

What N tells us

- $N(x)\,dx$: droplets per cm^3 with $r^2 \in [x, x + dx]$
- Units cm$^{-3}\mu$m^{-2}: a *density* in size, like a histogram divided by its bin width
- Total concentration: $n = \int N\,dx$ (cm^{-3})
- Expected, chamber-averaged (not one sample)



What the solutions look like



Larger D_x gives a broader distribution, more large droplets, and fewer droplets in total (large droplets fall out quickly).

D_x (μm^4/s)	droplets per cm^3	mean radius
0.1	34,400	4.4 μm
1	29,900	4.8 μm
10	20,600	5.7 μm
100	10,900	7.9 μm

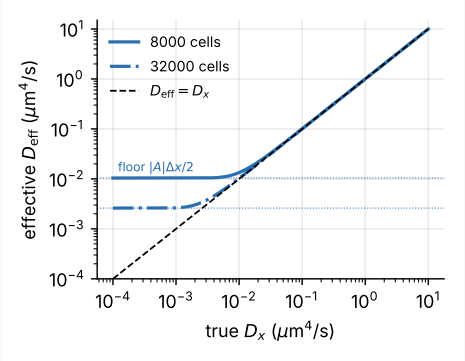
$s = 0.09\%$, NaCl $r_{\mathrm{dry}} = 0.065\,\mu\mathrm{m}$, $h = 1$ m, $F_{\mathrm{in}} = 100$ cm^{-3}s^{-1} (arbitrary scale).

Numerical solution

- Steve's finite-volume solver, in Python: 8000 cells in x; N_i = average of N over cell i
- Each cell balances flux in, flux out, and fallout (Scharfetter–Gummel fluxes, positive even when drift dominates):

$$\mathrm{A}(D_x)\,\mathbf{N} = \mathbf{b} \quad \text{(tridiagonal)}$$

- Reproduces the benchmark mean radii 4.22 and 5.74 μm ($D_x = 0.01,\ 10$)
- **Numerical diffusion:** the scheme smears like an extra $|A|\Delta x/2 \approx 0.01\,\mu\mathrm{m}^4/\mathrm{s}$, so small D_x needs $\Delta x \lesssim D_x/|A|$

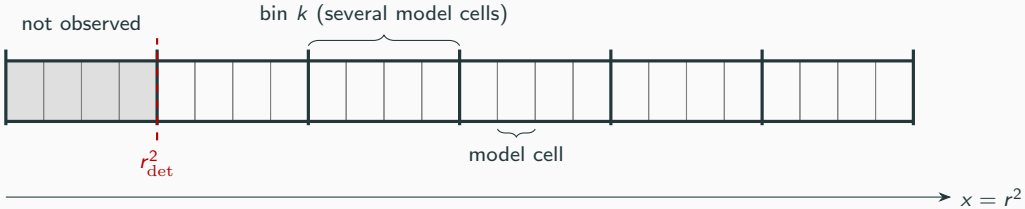


Below the floor, changing D_x no longer changes the computed solution.

Data and statistical model

What an instrument sees

- The instrument examines a small **sample volume** V of chamber air **(Q1)**
- It sorts the droplets it finds into size bins: data are **counts** c_k **(Q2)**
- It cannot see droplets below a **detection limit** r_{det} **(Q4)**



Bins below the detection limit are **unobserved, not zero**: there are plenty of droplets there; we just did not see them. They are left out of the fit. (Recording zeros would push D_x toward suppressing small droplets.)

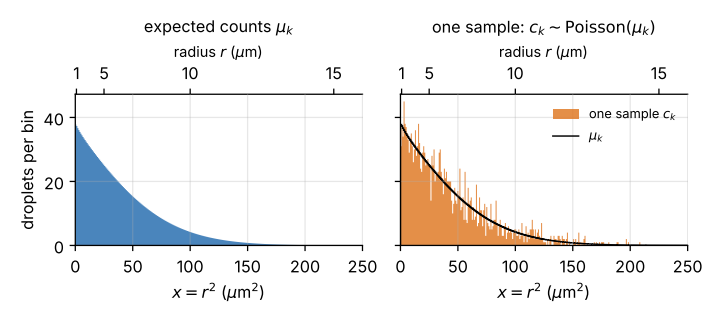
Expected counts μ_k and observed counts c_k

Well-mixed: every part of the chamber holds the same distribution on average, so

$$\mu_k = V \int_{\text{bin } k} N(x)\, dx$$

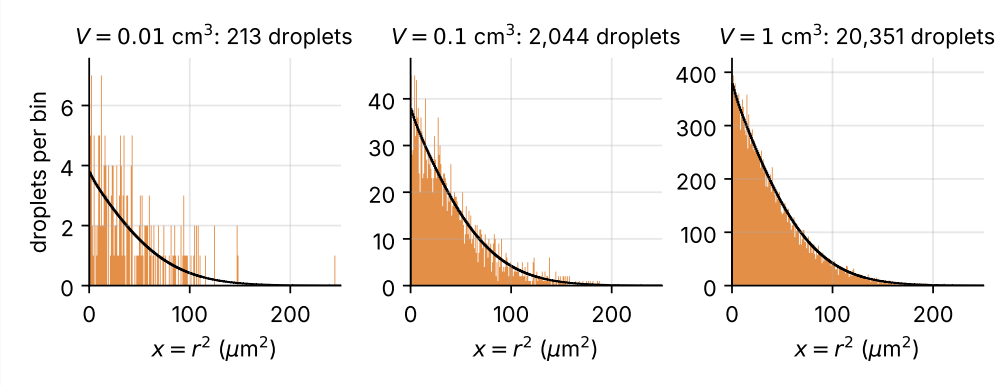
Each droplet is in the sample independently, with small probability $q = V/V_{\text{chamber}}$:

$$c_k \sim \text{Poisson}(\mu_k), \qquad \text{Var}\, c_k = \mu_k$$



$D_x = 10$, $V = 0.1$ cm^3 ($\approx$ 2000 droplets), bins of $0.9\,\mu\text{m}^2$. The busiest bin expects 38 droplets; relative noise is $1/\sqrt{\mu_k}$.

The number of droplets sets the noise



One sample (orange) at each volume against the expected counts (black). Well-mixed removes *bias*; it does not remove luck-of-the-draw *noise*.

Loss function: from likelihood to Poisson deviance

Independent Poisson counts give the likelihood of the observed histogram, a function of $\ell = \ln D_x$:

$$L(\ell) = \prod_{k \in \text{obs}} \frac{\mu_k(\ell)^{c_k}\, e^{-\mu_k(\ell)}}{c_k!} \qquad \Longrightarrow \qquad -\ln L(\ell) = \sum_{k \in \text{obs}} \Big[\mu_k - c_k \ln \mu_k + \ln c_k!\Big]$$

Three changes (none moves the minimum):

1. Drop $\ln c_k!$: it does not depend on D_x
2. Subtract its best possible value (at $\mu_k = c_k$), so a perfect fit scores 0
3. Multiply by 2

Poisson deviance

$$\mathcal{J}(\ell) = 2 \sum_{k \in \text{obs}} \Big[c_k \ln \frac{c_k}{\mu_k} \;-\; c_k \;+\; \mu_k \Big]$$

- For large counts, $\mathcal{J} \approx \sum_k (c_k - \mu_k)^2/\mu_k$: least squares with each bin weighted by its own noise
- Plain least squares would let the busy peak bins dominate and nearly ignore the sparse tail

Estimation

Deriving the adjoint gradient

The loss depends on D_x only through the model solution: $\mathcal{J} = \mathcal{J}\big(\mathbf{N}(D_x)\big)$, where $\mathrm{A}(D_x)\,\mathbf{N} = \mathbf{b}$.

1. Chain rule

$$\frac{d\mathcal{J}}{dD_x} = \mathbf{g}^\top \frac{d\mathbf{N}}{dD_x}, \qquad \mathbf{g} = \frac{\partial \mathcal{J}}{\partial \mathbf{N}}$$

2. Differentiate the model ($\mathbf{b}$ does not depend on D_x)

$$\frac{\partial \mathrm{A}}{\partial D_x}\mathbf{N} + \mathrm{A}\,\frac{d\mathbf{N}}{dD_x} = \mathbf{0} \;\Rightarrow\; \frac{d\mathbf{N}}{dD_x} = -\mathrm{A}^{-1}\frac{\partial \mathrm{A}}{\partial D_x}\mathbf{N}$$

Computing $d\mathbf{N}/dD_x$ directly costs one solve per parameter.

3. Substitute

$$\frac{d\mathcal{J}}{dD_x} = \underbrace{-\mathbf{g}^\top \mathrm{A}^{-1}}_{\boldsymbol{\psi}^\top}\,\frac{\partial \mathrm{A}}{\partial D_x}\mathbf{N}$$

4. Solve for $\boldsymbol{\psi}$ instead. The row vector $\boldsymbol{\psi}^\top = -\mathbf{g}^\top \mathrm{A}^{-1}$ does not depend on which parameter we differentiate:

$$\boldsymbol{\psi}^\top \mathrm{A} = -\mathbf{g}^\top \quad\Longleftrightarrow\quad \mathrm{A}^\top \boldsymbol{\psi} = -\mathbf{g}$$

5. Gradient

$$\frac{d\mathcal{J}}{dD_x} = \boldsymbol{\psi}^\top \frac{\partial \mathrm{A}}{\partial D_x}\mathbf{N}, \qquad \frac{d\mathcal{J}}{d\ell} = D_x\,\frac{d\mathcal{J}}{dD_x}$$

Ingredients

- From the loss, with $\boldsymbol{\mu} = V\mathrm{H}\mathbf{N}$: $\mathbf{g} = V\mathrm{H}^\top\big[2(1 - c_k/\mu_k)\big]$
- $\partial \mathrm{A}/\partial D_x$: differentiate the flux coefficients; same tridiagonal pattern as A

The adjoint in practice

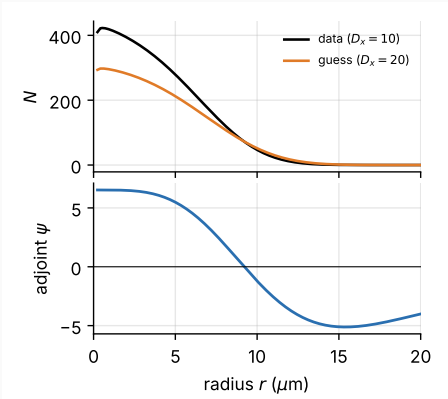
Each evaluation of loss and gradient

1. Forward solve $\mathrm{A}\mathbf{N} = \mathbf{b}$ (LU factorization)
2. Loss $\mathcal{J}$ and $\mathbf{g} = \partial\mathcal{J}/\partial\mathbf{N}$
3. Adjoint solve $\mathrm{A}^\top\boldsymbol{\psi} = -\mathbf{g}$, reusing the same LU factors
4. Gradient $\boldsymbol{\psi}^\top(\partial\mathrm{A}/\partial D_x)\mathbf{N}$

What $\boldsymbol{\psi}$ means. Adding a small source $\delta\mathbf{b}$ of droplets changes the loss by

$$\delta\mathcal{J} = -\boldsymbol{\psi}^\top\delta\mathbf{b},$$

so $\psi_i > 0$ where adding droplets would *improve* the fit.



At $D_x = 20$ the model is short of small and mid-sized droplets ($\psi > 0$) and has too many large ones ($\psi < 0$).

Why the adjoint?

Gradient method	cost for P parameters
Finite differences	$2P$ forward solves
Forward sensitivities	P extra solves
Adjoint	**1 extra solve, any P**

- The adjoint solve reuses the factorization of A, so it is cheap
- Exact gradient of the computed loss
- With one parameter it is not *needed*. It is an investment: s, injection rate, an unknown initial distribution, or a time-varying $s(t)$ add many unknowns but no extra cost

Optimizer

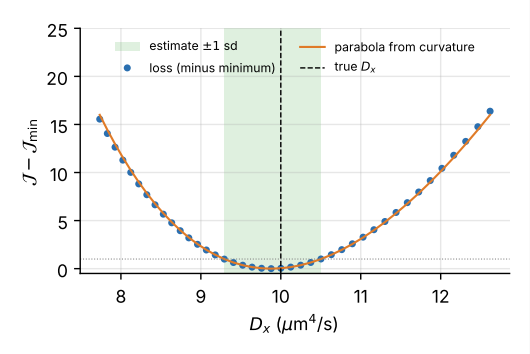
- Gradient descent in $\ell = \ln D_x$
- Step size from the last two gradients
- At most a factor e change per iteration
- Backtrack until the loss decreases

An error bar from a single sample

- Real data come as **one** sample, so we need an error bar from that sample alone
- Near its minimum the loss is a parabola, and $\mathcal{J} = -2 \ln L + \text{const}$, so the likelihood is Gaussian in ℓ:

$$\mathrm{sd}(\hat{\ell}) \approx \sqrt{2/\mathcal{J}''(\hat{\ell})}$$

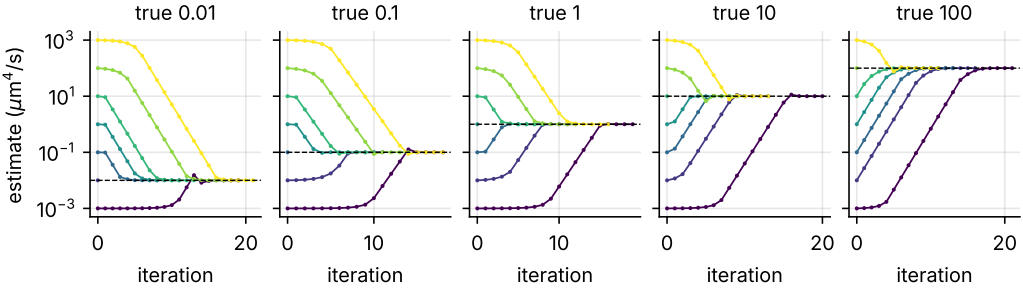
- $\mathcal{J}''$: finite difference of the adjoint gradient
- $\ell = \ln D_x$, so this is a *relative* error
- Predicted in advance by the Fisher information $\mathcal{I} = \sum_k (\partial_\ell \mu_k)^2 / \mu_k \propto V$



One sample, true $D_x = 10$, ≈ 2000 droplets:
$\hat{D}_x = 9.88$, sd $= 6\%$, interval $[9.3, 10.5]$.

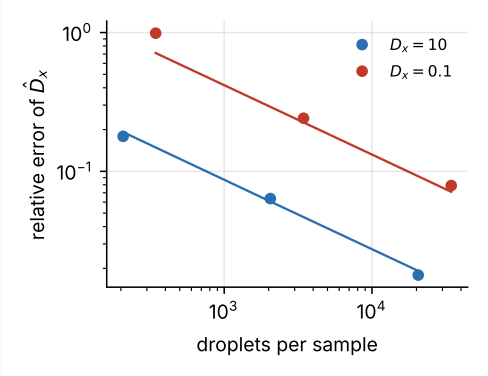
Results

Sanity check: exact data



- 5 true values $\times$ 7 starting guesses (10^{-3}, dark, to $10^3\,\mu\mathrm{m}^4/\mathrm{s}$, yellow): all recovered to a relative error $< 5 \times 10^{-9}$, in at most 21 iterations
- Starting guesses $\leq 10^{-4}$ fail: below the numerical-diffusion floor the gradient vanishes

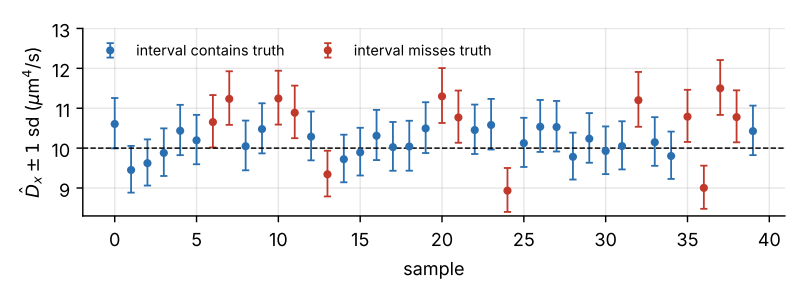
Precision versus number of droplets



Spread of $\hat{D}_x$ over 200 independent samples (dots) vs. prediction $1/\sqrt{\mathcal{I}}$ (lines).

- Estimates are unbiased
- Error $\propto 1/\sqrt{\text{droplets}}$. For $D_x = 10$: 19%, 6%, 2% with about 200, 2000, 20,000 droplets
- $D_x = 0.1$ is about $5\times$ harder for the same number of droplets
- With very little information (error $\gtrsim 50\%$) estimates skew low and the prediction is optimistic

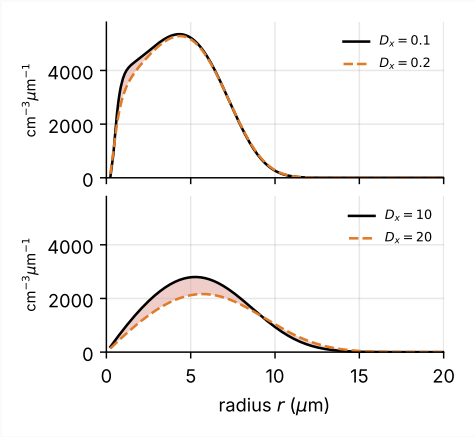
Single-sample error bars are honest



First 40 samples, true $D_x = 10$, about 2000 droplets each: 27 of 40 intervals contain the truth.

D_x	10			0.1		
droplets per sample	200	2000	20,000	340	3400	34,000
truth within ± 1 sd (ideal 68%)	71%	69%	72%	67%	62%	64%
truth within ± 2 sd (ideal 95%)	97%	93%	97%	95%	94%	91%

Why is large D_x easier to recover?

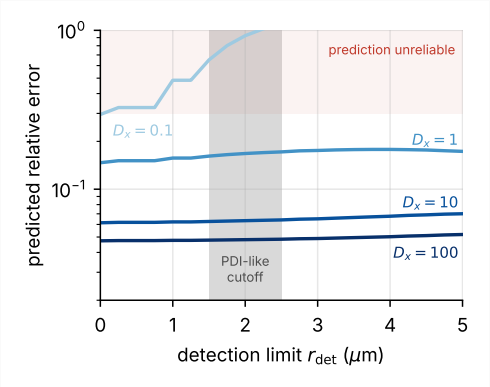


Doubling D_x changes the distribution by 4% at $D_x = 0.1$ (top), 22% at $D_x = 10$ (bottom).

- Not breadth itself: the **share of the width caused by** D_x
- Even with no turbulence the distribution is broad: droplets of many different ages coexist. D_x only adds to that
- Small D_x adds very little, so the data barely respond to it
- Physics, not numerics: information per droplet at $D_x = 0.1$ is unchanged on a $4\times$ finer grid
- Relative vs. absolute: with 2000 droplets, 0.1 ± 0.024 vs. 10 ± 0.6

D_x (μm^4/s)	0.1	1	10	100
share of width from D_x	2.5%	18%	48%	62%
pred. error (2000 drops)	24%	15%	6%	5%

Detection limit: harmless for large D_x, severe for small



Predicted error with 2000 droplets counted above the cutoff.

Monte Carlo, 200 samples each (relative error of $\hat{D}_x$):

cutoff r_{det}	$D_x = 10$ 2000 droplets	$D_x = 0.1$ 34,000 droplets
none	6.4%	7.9%
1 μm	6.0%	12%
2 μm	6.9%	27%
3 μm	7.0%	56%

- Large D_x: its fingerprint is spread over the whole distribution
- Small D_x: its fingerprint is near activation, $r \approx 1$–$2\,\mu$m, exactly what a cutoff hides

Takeaways and next steps

Takeaways and caveats

Takeaways

- Exact data: D_x recovered from any reasonable starting guess (above the grid's numerical-diffusion floor)
- Counts: unbiased estimates, error $\propto 1/\sqrt{\text{droplets}}$
- A single sample's curvature error bar is trustworthy
- Large D_x is easy; small D_x is hard because turbulence is a small share of the width
- The detection limit matters mainly for small D_x

Caveats

- *Inverse crime*: data generated by the same model and grid that we fit
- Poisson assumes independent sampling; real counts may vary more (clustering, a fluctuating chamber-wide s)
- D_x is *effective*: any unmodeled broadening (sizing error, fluctuating s, wall effects) gets absorbed into it
- The well-mixed assumption is weakest for large, fast-falling droplets

Next steps

1. **Drift and diffusion together:** estimate s with D_x. Both affect the width; s also sets the scale. Can a steady distribution separate them? (Note a and s enter as the product $a\,s$.)
2. **Remove the inverse crime:** generate the truth on a finer grid than the fit
3. **Stochastic particle model:** finite numbers of droplets with the same physics, as a stand-in for Lagrangian simulation data
4. **Simulation output:** C-ODT, then bin LES, super-droplet LES, DNS. Compare fitted D_x with $a^2\sigma_s^2\tau_s$ measured along droplet paths
5. **Chamber data**, then time dependence

Questions for the MTU group

Q1. What is the volume ratio $q = V/V_{\text{chamber}}$?

What we assume

- Poisson counts need each droplet to land in the sample independently, with $q \ll 1$
- Pi Chamber ≈ 3 m^3, samples of 0.01–1 cm^3: $q \approx 3 \times 10^{-9}$–$3 \times 10^{-7}$
- Simulations that report every droplet have $q = 1$: no sampling noise at all

What we'd like to know

- What effective volume is behind one histogram? (PDI: probe area $\times$ flow speed $\times$ averaging time?)
- Do successive histograms recount the same droplets? Droplets live for minutes.
- In simulations, how much does the whole-domain distribution fluctuate between snapshots of a steady run?

Q2. How many droplets are in a sample?

Predicted error in $\hat{D}_x$, detection limit 2.5 μm:

	droplets counted		
D_x (μm^4/s)	200	2000	20,000
100	15%	5%	1.5%
10	20%	6%	2%
1	54%	17%	5%
0.1	$>100\%$	$>100\%$	35%

From the Fisher information; reliable below about 30%.

Our setup: the model's concentration ($\approx 2 \times 10^4$ cm^{-3}) comes from an arbitrary injection rate.

What we'd like to know

- Typical droplet concentration, and droplets per histogram
- Averaging time per histogram; how many independent histograms per steady state
- A realistic activation rate for the model

Q3. What are realistic values of D_x?

Our reading of the stochastic theory (Chandrakar et al. 2016, 2018):

$$D_x = a^2 \sigma_s^2 \tau_s = \frac{a^2\, \sigma_{s_0}^2\, \tau_t}{(1+\mathrm{Da})^2}, \qquad a = 2\xi$$

With their fitted $\tau_t \approx 40$–48 s and σ_{s_0}, in $\mu\mathrm{m}^4/\mathrm{s}$:

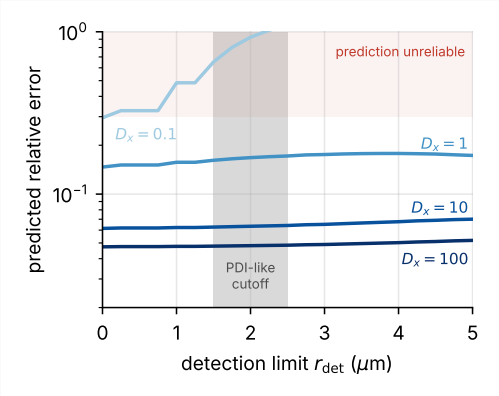
	Da = 0.3	1	5	40
$\Delta T = 16$ K	40	18	2	–
$\Delta T = 19$ K	260	110	12	0.3

$\Rightarrow$ roughly 1–200 $\mu\mathrm{m}^4/\mathrm{s}$.

What we'd like to know

- Is this mapping to D_x right?
- Can DNS/LES give supersaturation statistics along droplet paths (σ_s, τ_s), for a direct check?
- With $\tau_t \approx 40$–50 s and growth times of minutes, is a constant diffusivity adequate?

Q4. What is the detection limit?



Predicted error with 2000 droplets counted above the cutoff.

Chandrakar et al. (2018), phase Doppler: kept diameters $\geq 5\,\mu\mathrm{m}$, i.e. $r_{\mathrm{det}} \approx 2.5\,\mu\mathrm{m}$.

- $D_x \gtrsim 1$: the cutoff costs little
- $D_x \lesssim 0.1$: its fingerprint is near activation (1–2 μm) and is lost

What we'd like to know

- Which instruments and cutoffs apply now?
- Detection efficiency vs. size near the cutoff?
- Sizing uncertainty and bin edges?

Q5 and Q6. What else shapes the measured distribution?

Q5. What else broadens it?

Anything missing from the model is absorbed into the fitted D_x:

- Instrument sizing error
- Slow fluctuations of the chamber-wide s (large-scale circulation, oscillation)
- Subsaturated wall regions; large droplets concentrated near the floor
- Polydisperse aerosol; coalescence in the tail

Which of these matter most in the chamber, and how large are they?

Q6. Can the mean supersaturation be measured?

- s also affects the width (higher s: droplets grow larger before falling out), so a wrong s biases D_x
- Our next step estimates s and D_x together
- Is s known from temperature and humidity measurements, diagnosed from LES, or must it be inferred?
- How steady is s over the time one histogram takes?

Summary of asks

1. **Volume ratio q:** effective volume per histogram; are successive histograms independent?
2. **Droplets per sample:** concentrations, droplets per histogram, averaging time
3. **Realistic D_x:** is $D_x = a^2\sigma_s^2\tau_s$ right; supersaturation statistics along droplet paths
4. **Detection limit:** cutoff, efficiency curve, sizing uncertainty
5. **Other broadening:** which unmodeled effects matter, and how much
6. **Mean supersaturation:** measured, diagnosed, or estimated?

Most useful simulation data: whole-domain size distributions (or droplet lists with multiplicities) from statistically steady runs, with the run configuration (ΔT, injection rate, aerosol size, chamber height).

Backup

Backup: why the counts are Poisson

A bin holding M droplets; each is in the sample independently with probability $q = V/V_{\text{chamber}}$:

$$P(c) = \binom{M}{c} q^c (1-q)^{M-c} = \underbrace{\frac{M \cdots (M-c+1)}{M^c}}_{\to 1} \frac{\mu^c}{c!} \underbrace{\left(1 - \frac{\mu}{M}\right)^M}_{\to e^{-\mu}} \underbrace{\left(1 - \frac{\mu}{M}\right)^{-c}}_{\to 1} \longrightarrow \frac{\mu^c e^{-\mu}}{c!}$$

as $M \to \infty$, $q \to 0$ with $\mu = Mq$ fixed.

- Bins are independent: a Poisson total split independently among bins gives independent Poisson bins
- Neighboring bins have similar *expected* counts (the model handles that); independence is about their *errors*
- Not covered: clustering, a fluctuating chamber-wide s, an uncertain V. These correlate bins and make counts more variable than Poisson.

Backup: the Scharfetter–Gummel flux

Between neighboring cell centers, treat A and F as constant. Then $D_x N' = AN - F$ is a linear ODE with solution $N = F/A + Ce^{A(x-x_i)/D_x}$. Matching N_i and N_{i+1} gives

$$F_{i+1/2} = \frac{D_x}{\Delta x}\Big[B(-\mathrm{Pe}_i)\, N_i - B(\mathrm{Pe}_i)\, N_{i+1}\Big], \qquad \mathrm{Pe}_i = \frac{A_{i+1/2}\Delta x}{D_x}, \quad B(z) = \frac{z}{e^z - 1}$$

- $|\mathrm{Pe}| \ll 1$: ordinary diffusion plus drift times the average of N_i, N_{i+1}
- $\mathrm{Pe} \gg 1$: $F \to A\, N_i$, the upstream value; never negative
- Equivalent to central differencing with $D_{\mathrm{eff}} = D_x \frac{\mathrm{Pe}}{2} \coth \frac{\mathrm{Pe}}{2} \geq D_x$; for large Pe, $D_{\mathrm{eff}} \to |A|\Delta x/2$, independent of D_x